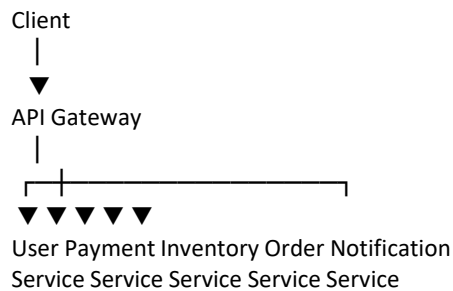


API Gateway

19 July 2026 03:41 PM

What is an API Gateway?

An **API Gateway** is a **single entry point** between clients (Web, Mobile Apps) and backend services (Microservices). Instead of calling multiple services directly, clients send requests to the **API Gateway**, which forwards them to the correct service.



Why Do We Need an API Gateway?

Without API Gateway ✗

- Client must know every service.
- Every service handles authentication.
- Hard to manage security.
- Duplicate code.

With API Gateway ☑

- Single endpoint for clients.
- Centralized security.
- Easier routing.
- Better scalability.

Core Features of API Gateway

Feature	Purpose
🔑 Authentication	Verify user identity
🛡️ Authorization	Check permissions
📊 Rate Limiting	Prevent abuse
⚖️ Load Balancing	Distribute traffic
⚡ Caching	Faster responses
🔄 Request Transformation	Convert request/response formats
🔍 Service Discovery	Find available services
🚫 Circuit Breaker	Stop sending requests to failed services
📊 Logging & Monitoring	Track requests and errors

1. Authentication

Who are you?

Checks identity using:

- API Keys
- JWT
- OAuth
- Certificates

2. Authorization

What are you allowed to do?

Checks permissions after authentication.

Example:

- Read Orders ☑
- Delete Orders ✗

3. Rate Limiting

Limits requests.

Example:

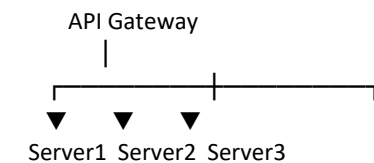
100 requests/minute

If exceeded:

429 Too Many Requests

4. Load Balancing

Distributes traffic.



Benefits:

- Prevents overload
- High availability

5. Caching

Stores frequently requested data.

Request



Cache



Found?

Yes → Return immediately

No → Database

Benefits:

- Faster
- Less database load

6. Request Transformation

Converts request formats.

Example

Client sends

Address

Gateway converts

GPS Coordinates

before sending to Delivery Service.

7. Service Discovery

Automatically finds healthy service instances.

Instead of

Payment Server = 192.168.1.5

Gateway asks

Where is Payment Service?

Service Registry replies

Payment Service



Server 2

8. Circuit Breaker

If a service keeps failing:

Gateway



Payment Service ✖



Stop sending requests temporarily

Prevents system-wide failures.

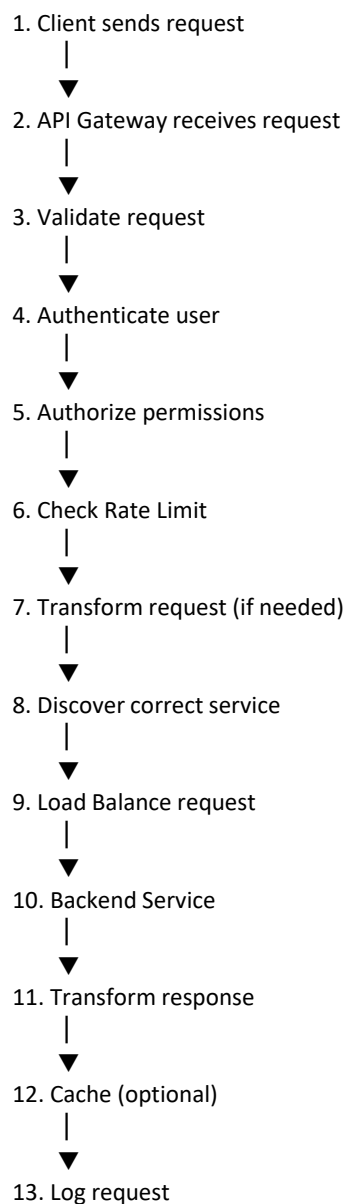
9. Logging & Monitoring

Gateway records:

- Request ID
- User
- URL
- Method
- Status Code
- Response Time
- Errors

Helps debugging and performance monitoring.

How API Gateway Works (Flow)





14. Send response to client